



Yeshwantrao Chavan College of Engineering

Hingna Road, Wanadongri, Nagpur-441110

(An Autonomous Institution affiliated to Yashwantrao Chavan Maharaaj Nagpur University)



Ph.: 07104-237919, 234623, 329249, 329250 Fax: 07104-232376, Website: www.ycce.edu

Department of Computer Technology

Vision Statement of the Department.

To be a preferred hub for nurturing computer engineering aspirants through innovative, ethical, value-based technical education, impactful research, and industry collaboration, to become successful professionals contributing for the betterment of society.

Mission Statement of the Department.

To nurture computer engineering professionals by offering Dynamic learning environment that encourages problem-solving using emerging technology, National Education Policy aligned evidence-based technical education, fostering innovation, research, and industry-institute collaboration & Holistic, ethical and sustainable development for lifelong success.

Session 2026-2027

Vision: Dream of where you want.

Mission: Means to achieve Vision

Program Educational Objectives of the program (PEO): (broad statements that describe the professional and career accomplishments)

PEO1	Preparation	P: Preparation	Pep-CL abbreviation pronounce as Pep-si-IL easy to recall
PEO2	Core Competence	E: Environment (Learning Environment)	
PEO3	Breadth	P: Professionalism	
PEO4	Professionalism	C: Core Competence	
PEO5	Learning Environment	L: Breadth (Learning in diverse areas)	

Program Outcomes (PO): (statements that describe what a student should be able to do and know by the end of a program)

Keywords of POs:

Engineering knowledge, Problem analysis, Design/development of solutions, Conduct Investigations of Complex Problems, Engineering Tool Usage, The Engineer and The World, Ethics, Individual and Collaborative Team work, Communication, Project Management and Finance, Life-Long Learning

PSO Keywords: Cutting edge technologies, Research

“I am an engineer, and I know how to apply engineering knowledge to investigate, analyse and design solutions to complex problems using tools for entire world following all ethics in a collaborative way with proper management skills throughout my life.” to contribute to the development of cutting-edge technologies and Research.

Integrity: I will adhere to the Laboratory Code of Conduct and ethics in its entirety.

Name and Signature of Student and Date

(Signature and Date in Handwritten)



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Session	2026-27(Odd)	CourseName	JAVA STACK LAB
Semester	5	CourseCode	23CT1301
RollNo	38	Name ofStudent	Aditya Patil

PracticalNumber	
Course Outcome	1. Understand various internet technologies. 2. Design the web pages using HTML and CSS. 3. Implement the XML technology to store the data. 4. Develop the interactive web pages using JavaScript.
Aim	a) Implement inheritance and method overriding using Java.
Problem Definition	A) Develop a Java program to demonstrate Inheritance and Method Overriding for an employee salary management system. Create a superclass Employee with a method calculateSalary(). Create two subclasses: Manager Programmer Override the calculateSalary() method in each subclass to display different salary details. In the main() method: Create objects using the parent class reference. Call the overridden method for each employee type. Display the salary information of both employees.
Theory (100 words)	Inheritance is an important concept of Object-Oriented Programming (OOP) in Java. It allows one class to acquire the properties and methods of another class. The class whose properties are inherited is called the superclass or parent class, while the class that inherits them is called the subclass or child class. In this program, Employee is the superclass, and Manager and Programmer are subclasses.



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	Method overriding occurs when a subclass provides its own implementation of a method already defined in the superclass. It enables runtime polymorphism, where a parent class reference can refer to different child class objects and invoke the appropriate overridden method. Here, calculateSalary() is overridden by both Manager and Programmer to display their respective salary details. Using parent class references demonstrates dynamic method dispatch and makes the program flexible and reusable.
Procedure and Execution (100 Words)	Step for Implementation: 8. Create the superclass: Define an Employee class with a method calculateSalary() to display general employee salary information. 9. Create the Manager subclass: Extend the Employee class using the extends keyword. Override calculateSalary() to display the Manager's salary details. 10. Create the Programmer subclass: Extend the Employee class and override calculateSalary() to display the Programmer's salary details. 11. Implement method overriding: Use the same method name, calculateSalary(), with the @Override annotation in both subclasses. 12. Create parent references: In the main() method, declare Employee reference variables and assign Manager and Programmer objects to them. 13. Call the overridden methods: Invoke calculateSalary() using the parent class references. Java determines at runtime which subclass method should execute. 14. Display output: The program displays different salary information for the Manager and Programmer, demonstrating inheritance, method overriding, and runtime polymorphism. 15. Compile and execute: Save the program, compile it using javac, and run it using java.



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Code:

```
class Employee {
    int id; String name; float salary;

    Employee(int id, String name, float salary) {
        this.id = id; this.name = name; this.salary = salary;
    }

    void calculateSalary() {
        System.out.println(id + " " + name + " Salary: " + salary);
    }
}

class Manager extends Employee {
    Manager(int id, String name, float salary) {
        super(id, name, salary);
    }

    @Override
    void calculateSalary() {
        System.out.println(id + " " + name + " Total Salary: " + (salary + 10000));
    }
}

class Programmer extends Employee {
    Programmer(int id, String name, float salary) {
        super(id, name, salary);
    }

    @Override
    void calculateSalary() {
        System.out.println(id + " " + name + " Total Salary: " + (salary + 5000));
    }
}

public class EmployeeInfo {
    public static void main(String[] args) {
        Employee e1 = new Manager(101, "ABC", 50000);
        Employee e2 = new Programmer(102, "XYZ", 30000);

        e1.calculateSalary();
    }
}
```



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	<pre>e2.calculateSalary(); } }</pre> Output: <pre>101 ABC Total Salary: 60000.0 102 XYZ Total Salary: 35000.0</pre>
Output Analysis	The program creates two objects using parent class (Employee) references. The first reference, e1, points to a Manager object with a basic salary of 50,000. When e1.calculateSalary() is called, the overridden method in the Manager class executes and adds 10,000 to the basic salary. Therefore, the total salary displayed is 60,000.0. The second reference, e2, points to a Programmer object with a basic salary of 30,000. When e2.calculateSalary() is called, the overridden method in the Programmer class executes and adds 5,000. Hence, the total salary displayed is 35,000.0. This output demonstrates inheritance, method overriding, and runtime polymorphism in Java. The appropriate subclass method is selected at runtime based on the actual object referred to by the parent class reference.
Link of student Github profile where lab assignment has been uploaded	
Conclusion	The Java program successfully demonstrates inheritance, method overriding, and runtime polymorphism. The Employee class acts as the superclass, while Manager and Programmer inherit from it



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	and provide their own implementations of the calculateSalary() method. By using Employee references to create subclass objects, the program shows how Java selects the appropriate overridden method at runtime. The different salary calculations for the Manager and Programmer confirm that method overriding provides flexibility and allows each employee type to have its own salary calculation. Thus, the program effectively demonstrates the practical use of important Object-Oriented Programming concepts in Java.
Plag Report (Similarity index < 12%)	8%
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